

# Project Report: Water Level Measurement and Automatic Tank Fill System

## 1. Overview

- **1.1. Introduction:**

- This project implements an automated water level monitoring and control system for tanks using an ultrasonic sensor and an STM32 microcontroller.
- It addresses the need for convenient and reliable water level monitoring, enabling automatic tank refilling to maintain a desired water level.
- The system is designed for ease of use and provides real-time water level information via a display.

- **1.2. Functionality Summary:**

- The system uses an ultrasonic sensor to measure the distance to the water surface.
- An STM32 microcontroller processes the sensor data, controls a relay to activate/deactivate a pump, and displays the water level on a display via I2C.
- The pump is automatically activated when the water level drops below 100cm and deactivated when it reaches 10cm.
- Accurate delays are implemented using timers and SysTick for precise timing of sensor readings and pump control.

## 2. Components and Hardware Setup

- **2.1. Hardware Components:**

- STM32 Microcontroller (STM32F446re)
- HC-SR04 Ultrasonic Sensor
- Relay Module
- Water Pump
- I2C Display (e.g., 16x2 LCD with I2C backpack)
- Power Supply
- Connecting Wires

- **2.2. Hardware Connections:**

- Ultrasonic Trigger (Trig): PA9 (Output)
- Ultrasonic Echo (Echo): PA8 (Input)
- Relay Input: PA5 (Output)
- I2C SCL: PB6
- I2C SDA: PB7
- Power connections for all modules.

### 3. Software Implementation

- **3.1. Development Environment:**
  - STM32CubeIDE
  - HAL libraries for STM32
- **3.2. Code Structure:**
  - main.c: Contains the main loop, sensor reading, pump control, and display update logic.
  - Timer functions for delay generation.
  - Interrupt handler for Echo pin.
  - I2C functions for display communication.
- **3.3. Key Functions and Algorithms:**
  - **Ultrasonic Measurement:**
    - Trigger pin is set HIGH for 10 microseconds to initiate a measurement.
    - Echo pin rising edge triggers an interrupt, capturing the start time.
    - Echo pin falling edge captures the end time.
    - Distance is calculated using the time difference and the speed of sound.
  - **Pump Control:**
    - If distance > 100cm, relay is activated (pump ON).
    - If distance < 10cm, relay is deactivated (pump OFF).
  - **Display Update:**
    - I2C communication is used to send the distance value to the display.
  - **Delay Generation:**
    - Timers are used for accurate microsecond delays for the ultrasonic trigger.
    - SysTick is used for longer delays when needed..

### 4. Conclusion

- **Summary of Achievements:**
  - Successfully implemented an automated water level monitoring and control system.
  - Achieved accurate distance measurement using an ultrasonic sensor.
  - Implemented reliable pump control using a relay.
  - Implemented stable I2C display communication.